

A Fixed-Domain Exhaustion Theorem

Standing, Consequence, and Compositional Closure

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Publication reference edition, 21 September 2026

Abstract

Non-degenerate determinate construction necessarily incurs admissibility, standing, reference, and irreversibility. This paper establishes a fixed-domain exhaustion theorem for its standing-bearing uses. The original bearer, questions, consequence relations, and continuation obligations are specified independently of a proposed realization. Their complete outcome profile yields a canonical quotient. Every proposed complete realization then either preserves those original requirements or has a witnessed defect: an incorrect or missing result, a collision of indistinguishable inputs with different required outcomes, an unsafe continuation or omitted safe region, or a failed premise, inference, or interpretation in an offered certificate. These alternatives quantify over interpreted proposals without requiring their correctness as an entrance condition.

The central compositional result derives preservation of every finite generated use from exact primitive conditions on observations, operation domains, and successor classes. It separates semantic coverage of the original task from completeness of an implementation. Sound complete classifiers for one original question are unique; the complete descriptive interface reduces to a single qualification bit precisely when all required outcomes are constant on that bit's fibres. Consequence determination fixes the unique realization of an original result, while a separate exact criterion establishes uniqueness across a family of candidate occupants.

History factorization, irreversible retention, and grounded authorization establish conservation under continuation and composition. Joint constructor descent extends the coverage argument to finite assemblies with relational applicability conditions. A short classification of task and profile changes records how a proposed variation affects the original requirements; the separate ATS development addresses component-role factorization and operator identity. The resulting impossibility claims concern the original task and survive faithful redescription. Worked finite and typed-construction examples expose both successful realizations and failed claims. An explicit export map makes the theorem usable as dependency support without transferring unstated domain facts or replacing proof of realization by a declaration of scope.

Keywords: exhaustion; admissibility; standing; observational equivalence; compositionality; canonical quotient; irreversible commitment; semantic coverage.

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1 The task of an exhaustion theorem

A construction may be presented in many ways while making the same claim about the same object. A proof may be renamed, a measurement record reformatted, or an operator implemented by a different sequence of steps. An exhaustion theorem must explain why these changes cannot supply an otherwise missing warrant, conceal a failed required use, or make two incompatible answers correct for one original incidence. It must also retain genuine new constructions and information that the original task actually requires. The purpose of this paper is to establish that closure in a form that subsequent mathematical and physical arguments can cite.

The kernel roles are necessary conditions of determinate construction itself [6]. Reference keeps a use attached to its bearer and question; admissibility concerns its actual conditions; standing concerns its eligibility for the specified use; irreversibility retains the identity and content of an original fixation. These roles are already incurred by determinate physical objects, states, carriers, records, and endpoints. A particular mathematical interface represents some of their uses. Proving that this interface is complete is an obligation concerning the representation, not an additional decision whether the objects are kernel-governed.

1.1 A familiar case: a bridge and its record

Consider a small bridge tested with a stated load and no outside support. The original question is whether that particular trial met those conditions. A photograph found later may establish that it did. A repaired bridge passing a later trial is a new successful occurrence. A finger supporting the bridge changes the conditions. None of these cases is settled merely by a label saying “pass.”

Suppose the intended use also includes repeating the trial. Two configurations may both pass initially while only one survives the repeat. Their current qualification has the same binary value, but their complete required-use profiles differ. A reader that retains only the initial bit cannot answer both repeat questions correctly. A reader that retains the necessary distinction still has to use it correctly. A list containing only one of two safe configurations can be sound while failing a claim to list all of them. These are different defects, with different witnesses.

The same distinctions arise in mathematics. Two well-typed terms need not have the same value. A successful reduction does not prove that its original term was well typed. A correct proof of a special case does not establish all cases. A quotient that preserves one judgment may discard a different judgment required for later composition. An exhaustion theorem must locate the failure without conflating the original object, its qualification, and the information needed for its further uses.

1.2 The theorem chain

The argument has seven stages.

1. **Original semantics.** Fix the original bearer, required questions, and their actual result relations before assessing a proposal. Different amounts of evidence do not define different original truths.
2. **Complete profile.** Construct the profile of every required outcome, including definedness where relevant. Derive its canonical quotient and the exact criterion for lossless representation.

3. **Compositional coverage.** Prove which primitive conditions preserve all finite generated uses. Establish the interpretation of the original uses in that language separately.
4. **Exact exhaustion.** Classify wrong and incomplete result graphs, deficient interfaces, and claimed complete interiors by explicit semantic witnesses.
5. **Conservation and authority.** Prove agreement for the same original incidence, retention of past outcomes, and soundness of grounded certificate composition.
6. **Consequence determination.** Separate uniqueness of a correct answer at one original input from a singleton output across a candidate family and from a unique occupant. State the exact additional conditions for those conclusions.
7. **Dependency export.** Combine these results into same-task impossibility and consequence-determination theorems, with explicit application conditions.

The principal result is fixed-domain exhaustion in Theorem 5.9, supported by the generated-use and joint-assembly proofs in Section 4. The complete-profile alternative in Theorem 3.3 supplies the information test on which those proofs rely. Theorem 8.1 and Corollary 8.2 remain explicit safeguards for downstream arguments: one output across a candidate family and one original occupant have different proof conditions.

1.3 What fixed domain fixes

A fixed domain fixes the original problem, its interpretation, and the uses for which completeness is claimed. It does not require every candidate implementation to be correct before the theorem can examine it. A false classifier can therefore be a candidate answer to the unchanged question. Its failure is a result of the theorem’s test, rather than a reason to exclude it from scrutiny.

A faithful change of coordinates may preserve the domain through an explicit interpretation. A new measurement may add evidence about an old occurrence. A new operation can produce a new state within an already declared repertoire. A change to the original question or repertoire is a change of task unless a transfer theorem connects it to the original one. The paper proves what such an interpretation must preserve. The phrase “same scope” abbreviates that semantic accountability; it supplies no correctness certificate by itself.

1.4 Literature and relation to the foundational papers

The future-behavior quotient belongs to the mathematical tradition of behavioral equivalence and coalgebraic systems [1]. Greatest invariant sets use fixed-point structure [2]. Their finite bad-path witnesses concern safety; eventual success requires additional semantics, as the safety/liveness distinction makes clear [3]. Standard preservation results for typed terms provide a concrete instance of independently justified safe reuse [4, 5]. These mathematical ingredients are used with local proofs; no priority claim is made for quotient factorization, induction, or fixed-point principles individually.

The foundational papers have the following division of labor.

1. **Kernel:** admissibility, standing, reference, and irreversibility as necessary work of non-degenerate determinate construction [6].

2. **AMetric Boundary:** content, symmetry, and non-transmission before and across the relevant fixation boundary [7].
3. **Structure of Admissibility:** complete interfaces, bivalent qualification, and the greatest admissible interior [8].
4. **Fixed-Domain Exhaustion:** coverage of original requirements, composition of generated uses and joint assemblies, and witnessed exhaustion of claimed realizations.
5. **Anchor, Tensor, and Skin:** the separate development of role factorization, component typing and minimality, and operator identity [12].

The dependency direction is explicit. This paper reuses the profile and interior results of *Structure of Admissibility*, states the forms required for its witnesses, and retains short local proofs for checkability. Its additional composition and coverage clauses are proved in Section 4. The ATS development can apply those general results to role-specific constructions; no component-factorization or operator-identity theorem from ATS is a premise of the exhaustion proof. Section 7 supplies the local task/profile effect split needed to describe a variation. The exhaustion proof depends on the stated original semantics, coverage, and realization conditions.

1.5 How the claims can fail

The mathematical claims are exposed to ordinary proof checking. A purported complete profile can omit a required question. A claimed decoder can encounter equal encoded inputs with different required answers. A proposed complete interior can omit a sound region or include an unsafe point. A role classification can confuse different variation domains. A proof can contain a false premise, invalid inference, or unproved interpretation. Each is a legitimate challenge without inventing an alternative physical regime.

Conversely, a faithful relabeling that preserves all original outcomes confirms the invariance conclusion. A separately warranted new construction is assessed at its own original incidence. The exhaustion theorem is non-generative about substantive domain facts: it neither selects a physical law nor produces a numerical constant merely from the names of its roles. It does construct mathematical profiles, quotients, and proof obligations; finite instances can admit effective verification. The distinction is between deriving consequences of specified data and supplying those data without warrant.

2 Original questions, qualification, and complete realization

2.1 The original requirement frame

Definition 2.1 (Original question semantics). Let X be a set of evaluated presentations of original incidences. Let $\mathcal{Q}$ be a set of original required questions. Each $q \in \mathcal{Q}$ has a value set V_q and an independently specified single-valued relation

$$G_q \subseteq X \times V_q.$$

Write $D_q = \text{dom}(G_q)$ and $f_q : D_q \rightarrow V_q$ for its associated partial function. The question identity includes its bearer, conditions, occurrence, and required use. A proposal's claimed answers are not used to define G_q .

The independence condition is semantic. For a bridge trial, the load, support conditions, and actual trial fix the question before its report is assessed. In a calculus, the syntax and original typing relation are fixed before a checker is tested. A domain may specify its result by mathematical conditions whose values are not presently known. This does not turn the proposal under assessment into its own standard of correctness.

Many-sorted inputs can be placed in a tagged disjoint union X ; each question is restricted to its appropriate inputs. Multi-valued specifications are also covered: if $H \subseteq X \times Y$ describes permitted outputs, use the Boolean membership questions $(x, y) \in H$. This retains the whole permitted set without silently choosing one occupant. A question about whether there exists a permitted output is a further Boolean question, whose answer does not identify that output.

Introduce a distinct symbol $\perp_q \notin V_q$ and define

$$O_q = V_q \sqcup \{\perp_q\}, \quad a_q(x) = \begin{cases} f_q(x), & x \in D_q, \\ \perp_q, & x \notin D_q. \end{cases}$$

This totalization represents semantic definedness. An uncomputed answer, missing evidence, or a reader's refusal to answer is not thereby the semantic outcome $\perp_q$. An explicit absence assertion must be evaluated against $X \setminus D_q$. An ordinary negative answer $0 \in \{0, 1\}$ is distinct from absence.

Definition 2.2 (Proposal, soundness, and coverage). For each q , a proposal supplies an interpreted graph $K_q \subseteq X \times V_q$ and a declared coverage domain $M_q \subseteq X$. Soundness requires $K_q \subseteq G_q$. Complete coverage on M_q requires

$$G_q \cap (M_q \times V_q) \subseteq K_q.$$

An explicit complete-outcome reader is instead compared with

$$G_q^\perp = G_q \cup \{(x, \perp_q) : x \notin D_q\}.$$

The full original-use problem takes the coverage domains required by that problem; a proposal cannot discharge it by quietly shrinking them.

Graphs allow inconsistent or multi-output proposals to be examined. A certificate reporting only some true positives can be sound. An assertion that it reports all positives adds the reverse inclusion. A total Boolean classifier makes negative assertions as well as positive ones, and its false negatives are incorrect original answers. These distinctions are important for exhaustion: absence of false positives is not completeness, and absence of proof is not a false original answer.

Theorem 2.3 (Original-incidence agreement and complete realization uniqueness). *Two sound readers for the same original question agree wherever they both answer. Any two sound readers with complete full coverage have exactly the graph G_q and hence coincide, including their semantic domains. The same conclusion holds for $G_q^\perp$ when complete outcome coverage is required.*

Proof. If (x, v) and (x, w) occur in sound graphs, both belong to the single-valued G_q , so $v = w$. Soundness and full coverage give both set inclusions between each graph and G_q . Equality follows. Totalization remains single-valued, so the argument also applies to $G_q^\perp$. $\square$

This proves uniqueness of a faithful complete realization of one original question. It does not assert equality of arbitrary candidate functions. If two candidates disagree, at least one assertion lacks the required original result; if a candidate answers too little, its coverage claim is separately assessed. No selector is needed to explain the mere mathematical existence of wrong candidates.

2.2 Qualification and necessary governance

A complete original eligibility question is represented by a predicate $A \subseteq X$ and its characteristic function $b : X \rightarrow \{0, 1\}$. Then $G_b = \{(x, b(x)) : x \in X\}$. Definedness of an actual partial reader remains a separate matter. Its unknown answer is not a third qualification value.

Corollary 2.4 (Unique qualification without collapse of the bearer). *For a fixed original qualification b , every sound complete Boolean classifier equals b . A qualification-preserving description can nevertheless retain distinctions between different points of $b^{-1}(1)$ or $b^{-1}(0)$. It represents the whole required interface by b alone exactly under the criterion of Theorem 3.6.*

Proof. The first statement is Theorem 2.3. Nothing in equality of a single coordinate equates the remaining original questions. The cited criterion proves when those coordinates are determined by it. $\square$

The qualification can be constant on a domain of determinate objects. This does not erase their other distinctions or remove kernel governance. Requiring both qualification values is needed for an assertion of exactly two realized qualification classes, not for the graph and interface theorems. Empty carriers and questions with empty semantic domains are handled by their stated set-theoretic conditions.

2.3 Same original task and faithful interpretation

Definition 2.5 (Faithful task transport). Suppose another presentation has carrier X' , questions $\mathcal{Q}'$, and outcome sets $O'_{q'}$. A full faithful transport consists of bijections $r : X \rightarrow X'$, $s : \mathcal{Q} \rightarrow \mathcal{Q}'$, and $v_q : O_q \rightarrow O'_{s(q)}$ preserving the absence markers, such that

$$a'_{s(q)}(r(x)) = v_q(a_q(x)) \quad (x \in X, q \in \mathcal{Q}).$$

Required coverage domains and any continuation/warrant data must also correspond when they belong to the original task.

Proposition 2.6 (Transport preserves original correctness and discrepancy). *Under a full faithful transport, soundness, complete coverage, and the existence of a surplus or omitted original record are invariant. A bijective recoding cannot remove a witnessed original discrepancy.*

Proof. The product bijection $(x, u) \mapsto (r(x), v_q(u))$ maps each totalized original graph onto its target graph and the candidate graph onto its interpreted recoding. A bijection preserves inclusion, equality, and both set differences. Restriction to the corresponding coverage domains gives the partial-coverage statement. $\square$

A useful one-way interpretation need not be bijective. It can prove a particular target result from a source result, but its implication and original meanings must then be stated. Section 6 handles such warrant-preserving composition. Calling the transport “lawful” is not a substitute for the displayed preservation equation; conversely, proving the equation establishes a genuine invariant independently of notation.

2.4 Conventions and proof obligations

The mathematical development uses classical set theory, set-sized question families, and ordinary function extensionality. Its semantic functions need not be computable. Universal assertions about all original uses require an interpretation of those uses in $\mathcal{Q}$. Section 4 supplies a compositional method for proving that interpretation for a generated use language. The canonical construction in the next section also applies directly to arbitrary specified families.

The proof obligations are separated deliberately: identity of the original task; coverage of its questions; soundness of a proposed answer; and completeness of the proposed realization. Kernel necessity does not make a defective representation complete, just as a physical object being determinate does not make every description of it correct.

3 Complete profiles and the canonical quotient

This section records the complete-profile and fibre principles used in *The Structure of Admissibility* [8], with explicit partial-result totalization and attained-image uniqueness for the exhaustion tests below. Their short proofs fix the exact forms used by the subsequent composition argument.

Definition 3.1 (Complete profile and observational quotient). For the original frame define

$$P : X \longrightarrow \prod_{q \in \mathcal{Q}} O_q, \quad P(x) = (a_q(x))_{q \in \mathcal{Q}}.$$

Put $x \equiv y$ exactly when $P(x) = P(y)$, and let $q_X : X \rightarrow \bar{X} = X/\equiv$ be the quotient map.

Theorem 3.2 (Canonical coverage and separation). *The profile exists for every specified original question family and recovers each original graph and its semantic domain. Distinct quotient classes have a separating original question. The map $[x] \mapsto P(x)$ is a bijection from $\bar{X}$ onto $P(X)$. The profile can be equivalently represented by Boolean membership coordinates without selecting a preferred output.*

Proof. Each single-valued partial relation gives the uniquely defined totalized a_q . Taking their product requires no selection among alternative values. The non- $\perp_q$ entries of coordinate q recover exactly G_q . Inequality of two profiles is inequality at some coordinate; therefore distinct classes are separated by an original question. The induced map to $P(X)$ is well defined by the equivalence definition, injective by its converse, and surjective by the attained-image definition.

For the Boolean representation take all coordinates (q, v) with $v \in O_q$, assigning 1 exactly when $a_q(x) = v$. Their equality implies equality of each original outcome, since its actual value has coordinate 1. Conversely equal original outcomes give equal Boolean coordinates. No choice of a single permitted occupant from a relational specification is made. $\square$

This is an explicit canonical coverage theorem for a specified original target, not an enumeration of familiar implementation types. Indexing by every original question prevents a favorable subfamily from being silently called complete. Whether a particular physical or mathematical inventory captures its intended target remains an application proof; an actual missing question is not repaired by the product notation.

Theorem 3.3 (Exact complete-interface alternative). *Let $h : X \rightarrow Z$ be any proposed representation. Exactly one of the following holds:*

1. There exists a unique decoder $d : h(X) \rightarrow P(X)$ such that $P = d \circ h$.
2. There exist $x, y \in X$ and $q \in \mathcal{Q}$ with $h(x) = h(y)$ but $a_q(x) \neq a_q(y)$.

Thus adequacy of h is equivalent to $\ker h \subseteq \ker P$. In the second case no function receiving only $h(x)$ can correctly return all original outcomes for every input.

Proof. A decoder makes equal encoded inputs have equal profiles. Conversely, if profiles are constant on every h -fibre, define $d(h(x)) = P(x)$. Constancy gives well-definedness, and the attained image gives both the codomain and uniqueness. Negating constancy produces x, y with unequal profiles; coordinate separation gives q . At their common encoded input a correct decoder would need both different outcomes, which a function cannot provide. $\square$

Existence of this semantic decoder does not certify a proposed decoder, its implementation, or its execution time. Such a proposal is tested against the original result graphs. The theorem also identifies why a renamed, randomized, or iteratively postprocessed code cannot deterministically recover a missing distinction from the same sole input. A randomized reader required to be correct with probability one at both indistinguishable inputs likewise cannot have two different almost-sure fixed outcomes under the same output law. An extra query can resolve the collision only by adding information; its own interpretation must be justified.

Corollary 3.4 (Canonical minimal information and unique minimal interface). *Every adequate h determines q_X uniquely. If h also identifies all points with equal complete profiles, then $h(X)$ and $\bar{X}$ are uniquely isomorphic over X . In particular, the minimal complete interface is unique up to the unique commuting bijection, while adequate richer representations may retain inert extra information.*

Proof. Compose the decoder of Theorem 3.3 with the inverse of the bijection in Theorem 3.2. The additional hypothesis is $\ker P \subseteq \ker h$, so the kernels are equal. The induced map between attained quotient sets is then a bijection. Surjectivity of their maps from X forces its uniqueness. $\square$

Theorem 3.5 (Fixed-function factorization and consequence determination). *For a specified $F : X \rightarrow Y$, the following are equivalent:*

1. F is constant on complete-profile classes;
2. there exists a unique $\bar{F} : \bar{X} \rightarrow Y$ with $F = \bar{F} \circ q_X$.

Every required totalized consequence a_q has this property. If F does not have it, a pair $x \equiv y$ with $F(x) \neq F(y)$ witnesses failure of factorization. Two sound complete realizations of the same original consequence agree, including after quotienting.

Proof. If (i) holds, set $\bar{F}([x]) = F(x)$; this is well defined and unique because q_X is onto. The converse follows by applying a function to equal quotient inputs. Failure gives the stated pair by negating constancy. Required consequences are coordinates of P . Their realization agreement follows from Theorem 2.3; equality of their induced maps then follows from surjectivity. $\square$

The theorem does not make every function on $\bar{X}$ equal. With $\bar{X} = \{0, 1\}$, both $z \mapsto z$ and $z \mapsto 1 - z$ are well-defined functions on the quotient. Only one can realize a specified original Boolean consequence at an input where their answers differ. Wrong but invariant consequences are

detected by original semantics, not by claiming they distinguish representatives within a fibre. The existence of a finite witness pair also differs from a finite procedure for verifying that the pair agrees on an infinite family of original questions.

Theorem 3.6 (Exact binary reduction). *Suppose a total qualification bit $b : X \rightarrow \{0, 1\}$ is among the original coordinates. Then the complete profile factors through b if and only if*

$$b(x) = b(y) \implies a_q(x) = a_q(y) \text{ for every } q \in \mathcal{Q}.$$

In that case $\ker b = \ker \mathbf{P}$ and $\bar{X} \cong b(X)$. There are exactly two realized complete classes if and only if both bit values occur. Constant qualification alone imposes no upper bound on the number of complete-profile classes.

Proof. The decoder criterion is Theorem 3.3 with $h = b$. Since b is a profile coordinate, profile equality always implies bit equality; the displayed converse gives equality of kernels. The attained-image construction proves the bijection and class count. For the last assertion take any nonempty X , the constant bit 1, and the additional identity observation $x \mapsto x$. The complete quotient then has exactly $|X|$ classes. $\square$

The result preserves bivalence of each original qualification without erasing identity or future-use distinctions. If base data $r : X \rightarrow B$ remain available, the corresponding relative criterion asks whether $\mathbf{P}$ factors through (r, b) . It does not require every original question to be determined by the bit in isolation. This is often the appropriate interface for a typed object whose identity and context are already retained.

Consumers of a complete interface. For any totalized original consumer f , including its definedness, insensitivity to equal-profile inputs is exactly factorization through the complete quotient, by Theorem 3.5. Naming that quotient does not constrain a separately chosen function on raw presentations. Requiring the accepted consumer to be a well-defined function on the quotient does. This is the general information-preservation condition used in the coverage argument; it does not require an ATS component classification.

4 Generated uses and compositional exhaustion

A complete semantic profile includes every question belonging to the original task. A second obligation arises when a practice presents its uses by formation rules: why do checks on its primitive observations and operations cover the uses formed from them? This section proves that coverage for finite deterministic construction. It retains the future-behaviour spine of exhaustion and separates it from the unsupported inference that one qualification bit determines every licensed consequence. The construction is related to behavioural equivalence for deterministic systems [1]; the application to admissibility uses the distinction between qualification and complete interfaces developed in [8]. All proofs needed here are supplied locally.

4.1 Partial operations and complete continuation profiles

Fix sets L of operation labels and J of observation labels. For $a \in L$ let $t_a : D_a \rightarrow X$ be a deterministic partial operation, with $D_a \subseteq X$. For $j \in J$ let $o_j : X \rightarrow V_j$ be a partial observation. Write

$$O_j = V_j \sqcup \{\perp_j\}, \quad a_j(x) = \begin{cases} o_j(x), & x \in \text{dom}(o_j), \\ \perp_j, & x \notin \text{dom}(o_j). \end{cases}$$

The displayed value in the first case is understood through the disjoint-union injection. Thus failure of definedness cannot be confused with a value. Operation labels a and the indexed outcome functions a_j have different types. No totality, finiteness of X , or finiteness of L or J is assumed.

For a set B , put $B^+ = B \sqcup \{\dagger_B\}$, with injection ι_B . This additional symbol will record an unavailable continuation; it is distinct from an available continuation whose observation is undefined. Let L^* be the finite words, including ε . Define $t_\varepsilon = \text{id}_X$ and $t_{aw} = t_w \circ t_a$ as a partial composite: the leftmost letter is performed first. Its domain is

$$D_\varepsilon = X, \quad D_{aw} = \{x \in D_a : t_a(x) \in D_w\}.$$

For $w \in L^*$ and $j \in J$ define

$$d_w(x) = 1_{D_w}(x), \tag{4.1}$$

$$A_{w,j}(x) = \begin{cases} \iota_{O_j}(a_j(t_w(x))), & x \in D_w, \\ \dagger_{O_j}, & x \notin D_w. \end{cases} \tag{4.2}$$

The continuation profile and its induced equivalence are

$$P_*(x) = ((d_w(x))_{w \in L^*}, (A_{w,j}(x))_{(w,j) \in L^* \times J}), \tag{4.3}$$

$$x \equiv_* y \iff P_*(x) = P_*(y). \tag{4.4}$$

Keeping the d_w coordinates separately makes the construction correct even when J is empty. It also states directly that the availability of a licensed use belongs to its behaviour.

Definition 4.1 (Observation-preserving congruence). An equivalence relation R on X is an observation-preserving congruence when, for every xRy , $a \in L$, and $j \in J$:

- (i) $x \in D_a$ if and only if $y \in D_a$;
- (ii) if $x, y \in D_a$, then $t_a(x)Rt_a(y)$;
- (iii) $a_j(x) = a_j(y)$.

Condition (i) is domain saturation. Condition (iii) includes both observation definedness and the observed value.

Theorem 4.2 (Largest continuation congruence). *The relation $\equiv_*$ is the largest observation-preserving congruence on X , where largest means inclusion as a subset of $X \times X$.*

Dependencies: Deterministic partial operations, disjoint totalizations, and all finite continuation coordinates in (4.4)..

Proof. Equality of profiles is an equivalence relation. If $x \equiv_* y$, equality of d_a gives domain saturation. Equality of $A_{\varepsilon,j}$ gives $a_j(x) = a_j(y)$ by injectivity of ι_{O_j} . Suppose $x, y \in D_a$. For every w , the definition of composition gives

$$d_w(t_a(x)) = d_{aw}(x), \quad A_{w,j}(t_a(x)) = A_{aw,j}(x),$$

and the same identities hold for y . The profile equalities at x, y therefore give every profile equality at $t_a(x), t_a(y)$. This proves congruence.

Conversely, let R be an observation-preserving congruence and let xRy . Induction on word length proves that $x \in D_w$ if and only if $y \in D_w$, and that, when defined, $t_w(x)Rt_w(y)$. The empty word is immediate. For aw , apply domain saturation and output congruence for a , then the induction hypothesis for w at the two resulting states. Thus all d_w agree. If the word is unavailable both $A_{w,j}$ values are $\dagger_{O_j}$; otherwise observation preservation at the resulting states makes them equal. Hence $x \equiv_* y$, so $R \subseteq \equiv_*$. $\square$

4.2 The exact primitive descent test

Let $h : X \rightarrow H$ be onto, with $H = h(X)$. Descent of a partial operation means descent of its domain as well as its output. In particular, a quotient may not declare an operation available on a class containing an unavailable original state.

Theorem 4.3 (Primitive and finite-word descent). *The following three conditions on h are equivalent.*

- (a) *Whenever $h(x) = h(y)$, primitive domains agree; primitive outputs satisfy $h(t_a(x)) = h(t_a(y))$ when defined; and $a_j(x) = a_j(y)$ for every j .*
- (b) *There are unique partial operations $\bar{t}_a : h(D_a) \rightarrow H$ and unique total observations $\bar{a}_j : H \rightarrow O_j$ such that*

$$h^{-1}(h(D_a)) = D_a, \quad \bar{t}_a \circ h|_{D_a} = h \circ t_a, \quad \bar{a}_j \circ h = a_j.$$

- (c) *Every finite word has a descended partial operation $\bar{t}_w : h(D_w) \rightarrow H$ with $h^{-1}(h(D_w)) = D_w$ and $\bar{t}_w \circ h|_{D_w} = h \circ t_w$; every a_j descends.*

When these conditions hold, $\ker(h) \subseteq \equiv_$. Consequently the map $p : X \rightarrow X/\equiv_*$ factors uniquely through h .*

Dependencies: Theorem 4.2; equality of h -fibres and surjectivity onto H .

Proof. Under (a), define $\bar{t}_a(h(x)) = h(t_a(x))$ for $x \in D_a$ and $\bar{a}_j(h(x)) = a_j(x)$. Output and observation agreement make the definitions independent of representatives. Domain saturation gives the inverse-image identity. These formulas also prove uniqueness. Conversely (b) gives each clause of (a) by evaluating its equalities at x, y in the same fibre.

To obtain (c) from (a), apply induction on words to the equivalence relation $\ker(h)$ exactly as in the second half of Theorem 4.2: domains of composites are saturated and the output h -classes agree. The same representative formulas then define $\bar{t}_w$. Taking the one-letter words in (c) gives the operation part of (b), and its last clause gives the observation part. Finally, (a) says that $\ker(h)$ is an observation-preserving congruence. Theorem 4.2 gives the inclusion. Define the factor by $h(x) \mapsto p(x)$; the inclusion proves well-definedness and surjectivity of h proves uniqueness. $\square$

The test has four direct witness forms. A same-fibre pair may disagree on an operation's availability; it may have defined outputs in different h -fibres; it may disagree on an observation's availability; or it may return different values of the same available observation. These are violations of the displayed equations, not new scope conventions. Checking only the last form misses domain loss and failure of compositional reuse.

4.3 A typed grammar of finite observations

We now make the formation rules explicit. An expression $e : B$ denotes a total function $\llbracket e \rrbracket : X \rightarrow B$. The grammar has the following constructors; every expression is a finite tree.

- (1) For $b \in B$, $\text{const}_b : B$ denotes the constant b .
- (2) For $j \in J$, $\text{obs}_j : O_j$ denotes a_j .
- (3) For $a \in L$, $\text{available}_a : \{0, 1\}$ denotes d_a .
- (4) If $e_i : B_i$ for $1 \leq i \leq n$ and $F : \prod_{i=1}^n B_i \rightarrow C$ is a fixed total function, then $F(e_1, \dots, e_n) : C$ denotes pointwise postprocessing by F . Here n is finite, including zero.
- (5) If $e : B$ and $a \in L$, then $\text{after}_a(e) : B^+$ denotes

$$x \mapsto \begin{cases} \iota_B(\llbracket e \rrbracket(t_a(x))), & x \in D_a, \\ \dagger_B, & x \notin D_a. \end{cases}$$

- (6) If $c : \{0, 1\}$ and $e_0, e_1 : B$, then $\text{if}(c; e_0, e_1) : B$ denotes $x \mapsto \llbracket e_{\llbracket c \rrbracket(x)} \rrbracket(x)$.

Postprocessing may be mathematically arbitrary, but its only inputs are the listed expression values. It cannot inspect x through an undeclared channel. Branching is deterministic and finite. Unbounded search, infinite-path success, and probabilistic outcomes are not silently added by these formation rules.

Lemma 4.4 (Continuation coordinates are generated). *For every w, j , the functions d_w and $A_{w,j}$ are denoted by expressions of the grammar.*

Proof. For the empty word use the constant 1 and postprocess obs_j by ι_{O_j} . Suppose d_w and $A_{w,j}$ have expressions. Apply after_a to the expression for d_w , then postprocess by $\{0, 1\}^+ \rightarrow \{0, 1\}$ sending $\iota_{\{0,1\}}(b)$ to b and $\dagger_{\{0,1\}}$ to 0. This gives d_{aw} . Apply after_a to the expression for $A_{w,j}$, then use the total function $(O_j^+)^+ \rightarrow O_j^+$ sending $\iota_{O_j^+}(z)$ to z and its outer failure symbol to $\dagger_{O_j}$. This gives $A_{aw,j}$. Word induction completes the construction. $\square$

Theorem 4.5 (Compositional coverage and the minimal complete interface). *For a surjection $h : X \rightarrow H$, the following are equivalent:*

- (a) Every expression $e : B$ has a unique factor $\bar{e} : H \rightarrow B$ with $\bar{e} \circ h = \llbracket e \rrbracket$.
- (b) All continuation coordinates $d_w, A_{w,j}$ factor through h .
- (c) $\ker(h) \subseteq \equiv_*$.

Thus $p : X \rightarrow X/\equiv_*$ is complete for exactly these generated observations, and it is the coarsest such interface. Every complete h admits a unique map $r : H \rightarrow X/\equiv_*$ with $r \circ h = p$. The primitive

conditions in Theorem 4.3 imply (a)–(c) and, in addition, permit every finite word to descend as a partial update on H .

Dependencies: Theorems 4.2 and 4.3 and lemma 4.4; the six finite formation rules above..

Proof. The lemma gives (a) $\Rightarrow$ (b). Agreement of all continuation coordinates on a fibre is exactly (c), so (b) $\Leftrightarrow$ (c), using representative independence and surjectivity for factorization.

For (c) $\Rightarrow$ (a), first prove by structural induction that every expression is constant on $\equiv_*$ -classes. Constants are immediate. Base observations and primitive availability are preserved by Theorem 4.2. Equal argument values give equal postprocessed values. For $\text{after}_a(e)$, congruence gives equal availability; if available, the two resulting states are $\equiv_*$ -equivalent, so the induction hypothesis gives equal child values; if unavailable both results are the same failure symbol. For a conditional, the induction hypothesis gives equal conditions and hence the same selected branch; its values agree by the induction hypothesis for that branch. This covers every constructor.

Now $h(x) = h(y)$ implies $x \equiv_* y$ by (c), so every expression is constant on h -fibres. Define $\bar{e}(h(x)) = \llbracket e \rrbracket(x)$. Well-definedness follows from the preceding equality, and uniqueness from surjectivity. Apply this result to p . The same representative argument defines $r(h(x)) = p(x)$ for any complete h , proving the asserted minimality. The final statement is Theorem 4.3 together with (c). $\square$

Remark 4.6 (Observation completeness and update descent are distinct). The converse of the last sentence is false if h retains distinctions that the generated observations do not require. For example, let $X = \{x, y, u, v\}$, let every base observation be constant, and let one total operation send x to u and y to v (fixing u, v). Every continuation profile is identical: all words are available and all observed values are constant. Choose $h(x) = h(y)$ but $h(u) \neq h(v)$. Every generated observation factors through h , yet the operation does not descend through h . The canonical interface p has no such surplus distinction and satisfies both results. A claim about reuse of a proposed enriched interface must therefore check output congruence, not only its observational completeness.

4.4 Finite mismatch witnesses for observations

Theorem 4.7 (Constructor exhaustion for a mismatch). *Suppose $h(x) = h(y)$ but an expression $e : B$ of the observation grammar has $\llbracket e \rrbracket(x) \neq \llbracket e \rrbracket(y)$. Then a finite descent through its expression tree identifies one of the primitive defects listed after Theorem 4.3, witnessed by a pair in one h -fibre. Conversely, if the primitive conditions hold, no such mismatch exists. Independently of those stronger conditions, failure of completeness for generated observations is equivalent to a same-fibre pair and a single finite continuation coordinate d_w or $A_{w,j}$ on which the pair disagrees.*

Dependencies: The finite grammar; Theorems 4.3 and 4.5..

Proof. Maintain a mismatching expression and a same-fibre pair. A constant cannot mismatch. At a base observation, unequal totalized values expose either different observation definedness or different defined values. At an availability expression, the pair is a domain-saturation defect. At postprocessing, unequal outputs imply that at least one of the finitely many arguments differs; continue into such an argument with the same pair. At a conditional, either its conditions differ, in which case continue into the condition, or the conditions agree and the selected branch differs, in which case continue into that branch.

At $\text{after}_a(e')$, different availability is a domain defect. If both operations are unavailable, mismatch is impossible. If both are available, their child values differ. When $h(t_a(x)) \neq h(t_a(y))$, the current pair witnesses output-congruence failure. Otherwise continue into e' with the same-fibre pair $t_a(x), t_a(y)$. Each continuation moves to a proper child of the finite tree, so this process terminates at a primitive defect. Its converse follows also from Theorems 4.3 and 4.5.

For the last assertion, failure of Theorem 4.5(c) supplies a same-fibre pair with unequal profiles. Inequality of indexed tuples supplies an index at which they differ. Such an index is either a finite word or a finite word together with an observation label. Conversely such a difference contradicts factorization of that coordinate, and Lemma 4.4 makes it a generated-observation mismatch. $\square$

The witness theorem exhausts the finite constructors. It is not a general algorithm for discovering a bad pair, deciding primitive semantic equality, or verifying an infinite family of operations. In particular, arbitrary postprocessing functions need not be computable. Finiteness concerns the particular expression or continuation witnessing a mismatch, not the size or decidability of the entire practice.

4.5 Joint compatibility and finite strict assembly

The observation grammar above uses unary partial updates on one carrier and returns totalized observation values. A many-sorted construction language can also combine several interpreted inputs through a partial constructor. Its joint applicability must be retained as well as its output. Checking each input separately does not establish compatibility of the tuple. The following criterion states that additional condition exactly; here J denotes the constructor's joint domain.

Theorem 4.8 (Exact descent of a typed constructor). *For $i = 1, \dots, n$ let $h_i : D_i \rightarrow H_i$ be retained component interfaces, and let $h_o : D_o \rightarrow H_o$ be the retained output interface. A declared constructor is a partial map*

$$C : J \longrightarrow D_o, \quad J \subseteq \prod_{i=1}^n D_i.$$

Write $h = \prod_i h_i$. There exists a unique partial constructor

$$\bar{C} : h(J) \longrightarrow H_o$$

with $h^{-1}(h(J)) = J$ and $\bar{C}(h(\vec{d})) = h_o(C(\vec{d}))$ for all $\vec{d} \in J$, if and only if:

1. equal retained input tuples have the same membership in J ;
2. for any two members of J with equal retained tuples, the retained outputs are equal.

Failure has a pair witness at a joint domain or output condition.

Proof. The displayed descended domain and function imply both conditions by applying equal inputs. Conversely, define $\bar{C}(h(\vec{d})) = h_o(C(\vec{d}))$ on $h(J)$. Condition (ii) gives well-definedness, and (i) gives the exact preimage domain. Every element of $h(J)$ has a representative in J , forcing uniqueness. Negating either condition gives the stated pair. For a nullary constructor the input product is a singleton and its own domain and output conditions remain explicit. $\square$

Take each retained interface to include the complete original profile and relevant target typing. A relation between components is then genuinely retained only if the joint domain test is saturated

on the product fibres. For example, if two component profiles are both constant but compatibility requires that their hidden identifiers be equal, the diagonal compatibility relation does not descend. Individually adequate descriptions of unary properties have not established adequacy for that relational use. The full joint question can instead be retained explicitly.

Theorem 4.9 (Finite assembly through retained interfaces). *Let a many-sorted description language consist of variables, constants, and finite constructor trees whose original interpretations are fixed. Evaluation is strict: a constructor application is undefined if any required child is undefined; otherwise its joint domain determines applicability. Suppose every constructor satisfies Theorem 4.8 for the retained interfaces. Then every generated description descends through the retained interfaces, including its definedness. Equal retained leaf data give equal definedness and equal retained result. If a generated term violates this property, a finite descent through its tree exposes a failed joint domain or output condition at a constructor instance.*

Proof. Proceed by induction on the tree. Variables transmit their retained inputs. A constant is a nullary constructor. At an internal node, if a child is unavailable the same child is unavailable for equal retained leaves by induction, so the composite is unavailable in both evaluations. Otherwise the children’s retained results agree; constructor-domain saturation gives equal applicability, and output descent gives equal retained results when applicable.

For a mismatch, inspect its root. If a child pair already differs in definedness or retained value, continue into that child. Otherwise all retained child data agree, so the root witnesses a joint domain or output failure. Each descent reaches a proper subtree; finiteness yields termination. Repeated use of a variable causes no difficulty because equal complete assignments give equal values at each occurrence. Any additional alias or resource condition must be in the relevant joint domain. $\square$

The theorem covers the strict finite constructor language just specified. To cover a practice, every original construction in that practice must have an interpretation-preserving expression in the language. Such a proof can be by induction on its independently specified formation rules. An operation not generated by those rules is not eliminated by the induction; if it belongs to the original target, its omission defeats the claimed coverage. Direct semantic profiles remain available when no finite grammar is claimed.

4.6 Coverage of the original target

Corollary 4.10 (Coverage of an independently fixed question family). *Let the original task consist of questions $q \in \mathcal{Q}$ with totalized answers $a_q : X \rightarrow O_q$. Suppose that for each original question there is a well-typed expression $e_q : O_q$ of the observation grammar and a proof of the interpretation identity*

$$\llbracket e_q \rrbracket = a_q.$$

Then every interface complete for that grammar is complete for the original question family. If, additionally, each generated expression is a fixed total postprocessing of finitely many original answer coordinates, then completeness for the original family and completeness for the grammar are equivalent.

Dependencies: The displayed interpretation identities; Theorem 4.5; for the converse, the stated finite-coordinate representation of each expression..

Proof. Factor each e_q through the interface and use its interpretation identity. For the converse, factor the finitely many original coordinates occurring in the given representation and compose their factors with the specified postprocessing function. This constructs the factor of every expression. $\square$

For strict assembly, Theorem 4.9 supplies descent of a term's definedness and retained result. An original readout from that result must itself factor through the retained output, and the composite totalized answer must be proved equal to a_q . Composing the descended term with that output reader then supplies the required answer factor. These interpretation and output conditions establish coverage of the original question; the constructor induction does not assert that every original construction belongs to its language.

The interpretation identity is an obligation to be established from the original use, not permission to redefine that use until a preferred grammar covers it. A loop with an eventual-success question, for example, is not covered merely because every finite execution prefix has an expression. Its success question must be represented and proved equal to the intended one, or retained directly as an additional semantic coordinate. For an arbitrary set-sized original family $\mathcal{Q}$, the complete semantic profile $\mathbf{P} = (a_q)_{q \in \mathcal{Q}}$ remains available without any finite-grammar claim. The present theorem supplies an additional compositional proof when the declared formation rules justify it. This distinction concerns coverage of particular uses; it introduces no optional condition on the kernel governance of determinate objects, states, records, or endpoints.

Corollary 4.11 (Total-update future behaviour). *If every primitive operation is total, then*

$$x \equiv_* y \iff a_j(t_w(x)) = a_j(t_w(y)) \text{ for every } w \in L^*, j \in J.$$

This is the largest forward-invariant equivalence preserving all present observations. For a single total update t and a single qualification observation $b : X \rightarrow \{0,1\}$, it is equality of the entire sequence $(b(t^n(x)))_{n \geq 0}$, not merely equality of $b(x)$.

Proof. All word domains are X , so all d_w equal 1. Remove the common outer injections from the remaining equalities and apply Theorem 4.2. With one update, the words are precisely its finite iterates. $\square$

Each answer coordinate may be bivalent while the full continuation profile has many classes. Exhaustion preserves every required distinction in that profile. Its conclusion is uniqueness of the canonical complete quotient up to the usual relabelling of classes, and impossibility of losing a required distinction while preserving every original answer. It is not a collapse of all future behaviour to one present qualification bit.

5 Exact exhaustion of results, interfaces, and continuations

An exhaustion proof has two duties: cover the original requirements and establish the alternatives for their realization. Theorem 3.2 supplies a complete semantic profile for the specified target family. Section 4 supplies a method for proving that a smaller, generated interface covers that family. We now classify failures against those independently interpreted requirements. This order prevents a list of failure names from serving as its own coverage premise.

5.1 Surplus and omission in original result graphs

Theorem 5.1 (Exact graph exhaustion). *For a proposed complete original graph K_q , exactly one of the following holds:*

1. $K_q = G_q$;
2. $K_q \setminus G_q \neq \emptyset$ and $G_q \setminus K_q = \emptyset$;
3. $G_q \setminus K_q \neq \emptyset$ and $K_q \setminus G_q = \emptyset$;
4. both differences are nonempty.

A surplus record is an incorrect original assertion. An omitted record defeats full coverage. For required coverage only on M_q , the omission test is restricted to $G_q \cap (M_q \times V_q)$; soundness still concerns all asserted records. The same alternatives apply to complete-outcome graphs $G_q^\perp$. Failure for any set-sized family of questions is witnessed at an original question and record.

Proof. Two sets are equal if and only if each is contained in the other. The emptiness or nonemptiness of their two directional differences gives exactly the four mutually exclusive alternatives. Membership in a nonempty difference supplies a record. Restricting the coverage side gives the stated partial case. If all questionwise conditions held, their conjunction would hold; therefore failure of the conjunction has a questionwise witness. The argument does not enumerate implementations. $\square$

A constant classifier can be representation-invariant and wrong. If it always reports 1 while the original qualification is 0 at x , $(x, 1)$ is surplus. Invariance cannot correct that result. If the true result is 1 and a total classifier says 0, it both supplies a false record and omits the true one. A sound partial positive certificate supplies neither false record merely by remaining silent.

5.2 The complete safe interior and its witnesses

Definition 5.2 (Independent continuation contract). Let $R \subseteq X \times L \times X$ be the declared labeled continuation relation, $A \subseteq X$ an original local qualification, and $W(x, \ell, y)$ an original edge-warrant predicate on R . A finite R -path is safe if every visited node lies in A and every traversed edge satisfies W . Include the length-zero path. Define

$$S = \{x : \text{every finite declared path starting at } x \text{ is safe}\},$$

$$\Phi(U) = \{x \in A : \forall (x, \ell, y) \in R, W(x, \ell, y) \text{ and } y \in U\}.$$

The relation and predicates are fixed before S is determined. Required operation availability, when part of the task, is included in A rather than inferred from the absence of edges.

The greatest-interior construction is the one developed in *The Structure of Admissibility* [8]. We retain a short proof here to expose the exact unsafe-path and omitted-region witnesses used in Theorem 5.9. The relation may be nondeterministic or infinitely branching. The universal quantifier expresses the promised repertoire: every declared continuation must preserve the stated conditions. An optional operation need not be available everywhere. A requirement that a particular operation be available has the form $\exists y (x, \ell, y) \in R$ and must be checked explicitly. Calling only already successful paths “declared” would change the original contract.

Theorem 5.3 (Unique greatest safe interior). *The operator Φ is monotone and*

$$S = \Phi(S) = \bigcup \{U \subseteq X : U \subseteq \Phi(U)\}.$$

Consequently every sound inductive region $U \subseteq \Phi(U)$ is contained in S . A sound region is complete for the original robust-use question if and only if it equals S . Every inclusion-maximal sound region equals S .

Proof. Monotonicity follows because enlarging U only relaxes successor membership. If $x \in S$, the zero-length path gives $x \in A$, and every one-edge path gives its warrant. Every path from a successor can be prefixed by that edge, so the successor lies in S . Thus $S \subseteq \Phi(S)$. Conversely, if $x \in \Phi(S)$, its zero-length path is safe, and every nonempty path starts with a warranted edge to a point of S , whose suffix is safe. Hence $\Phi(S) \subseteq S$.

For $U \subseteq \Phi(U)$, induction along any given finite path from U keeps its nodes in $U \subseteq A$ and verifies each edge. Thus $U \subseteq S$. Since S itself is such a region, it equals their union. A complete region must contain every actually robust point and hence contain S ; soundness gives the reverse inclusion. Any proper sound subset can be enlarged to the sound set S , proving the maximality statement. $\square$

The existence of S does not establish $S \neq \emptyset$. A nonempty sound inductive region proves nonempty realization, and conversely any point of S lies in the sound region S . Unique complete interior does not mean a unique occupant of that interior.

Theorem 5.4 (Exact interior exhaustion). *Let $I \subseteq X$ be a claimed complete robust interior. If $I \neq S$, it has surplus, omission, or both. Every $x \in I \setminus S$ starts a finite declared path with an unqualified node or an unwarranted edge. Every $x \in S \setminus I$ lies in a sound inductive region not contained in I . Conversely, either such witness defeats the corresponding soundness or completeness claim. In particular,*

$$I = S \iff I \subseteq \Phi(I) \text{ and } \forall U \subseteq X (U \subseteq \Phi(U) \Rightarrow U \subseteq I).$$

Proof. The two set differences cover inequality. The negation of membership in S is the existence of an unsafe finite path, which fails one of the independently fixed node or edge predicates. If $x \in S \setminus I$, take the sound region $U = S$; conversely any sound region is contained in S , so a point of it omitted by I is a completeness witness. The displayed conditions respectively give $I \subseteq S$ and, by taking $U = S$, $S \subseteq I$. Equality implies both conditions by Theorem 5.3. $\square$

Proposition 5.5 (Finite-depth exhaustion of unsafe use). *Put $U_0 = X$ and $U_{n+1} = \Phi(U_n)$. Then $U_{n+1} \subseteq U_n$ and $S = \bigcap_{n \geq 0} U_n$. On a finite carrier, at most $|X|$ strict removals occur before stabilization. No finite-branching hypothesis is needed.*

Proof. The first inclusion follows from $\Phi(X) \subseteq X$ and monotonicity. For any nonempty family (V_i) , direct expansion gives $\Phi(\bigcap_i V_i) = \bigcap_i \Phi(V_i)$: universal successor membership commutes with intersection, while A and W are fixed. Thus the descending intersection is a fixed point. It contains S by induction and is contained in S by greatestness. Every strict decrease on a finite carrier removes a point not removed again. Once equality occurs, application of Φ repeats it. $\square$

This provides finite path witnesses for universal local safety. An infinite path's every node and edge lies in a finite prefix, so the same local conditions hold along it. Eventual success or fairness is a different original question and may have no finite bad-prefix witness [3]. Such questions can be

included in the complete original profile, with their own semantics; the local-safety theorem is not their proof.

5.3 Certificate and interpretation defects

Definition 5.6 (Offered certificate obligation). An offered certificate is a well-founded dependency tree whose nodes carry interpreted original claims. Leaf warrants, rule licenses, and the interpretation of each transmitted premise are specified independently. Each use must satisfy its original warrant conditions. A certificate claiming valid support undertakes these conditions in addition to the truth of its conclusion. Where a condition is itself a determinate claim, its Boolean truth can be included among the original questions.

Lemma 5.7 (Localization of a false conclusion). *If an interpreted well-founded certificate has a false conclusion, then it has a false leaf or an inference instance with true immediate premises and a false conclusion. If a dependency silently changes the interpreted bearer or question, it instead fails to transmit the required premise unless an independent translation justifies that change.*

Proof. Choose a minimal false node in the well-founded dependency order. If it has no premises it is a false leaf. Otherwise every immediate premise is true by minimality while the conclusion is false. Such an instance does not preserve the original truths. The argument concerns actual interpreted premises. A different interpretation is a different premise and needs a valid implication to the required one. $\square$

The certificate can be defective even when its conclusion happens to be true. If a required license, premise warrant, or transfer condition is false, that original condition supplies the failed certificate witness. If its proof is merely missing, the warrant remains unestablished; a semantic failure has not thereby been proved. Section 6 shows how grounded rules can establish the conditions without a separate chronological inspection event.

5.4 The fixed-domain exhaustion theorem

Definition 5.8 (Interpreted realization obligation). Fix the original question family and its required coverage, together with any continuation contracts and certificate obligations belonging to the intended task. An interpreted proposal may claim: original answer graphs K_q ; recovery of complete outcomes from an interface h ; a complete robust interior I ; or warranted certificate support. A *successful complete realization* satisfies every obligation it undertakes and every original obligation required of the realization. Output meanings, coverage, and applicability are interpreted before correctness is tested.

Theorem 5.9 (Fixed-domain exhaustion with original witnesses). *For every independently specified original task of Definition 5.8, the complete profile of Theorem 3.2 covers its question family. Every interpreted proposal satisfies its original semantic obligations or has one or more of the following witnesses:*

1. **Original-result defect:** *a surplus asserted record or an omitted record in required coverage, including explicit absence claims when required.*
2. **Information defect:** *two equal encoded inputs with different required original outcomes, defeating every complete decoder from that interface.*

3. **Continuation defect:** a point in the claimed interior with a finite unsafe path, or an omitted point of an independently sound inductive region.
4. **Support defect:** a failed original premise, inference, license, or interpretation condition in offered support; a false certified conclusion has the local alternative of Lemma 5.7.

The witnesses are exhaustive for the specified obligations and may overlap. A proposal satisfying every original obligation has no further semantic defect relative to those same obligations. A reduced or generated interface establishes this full conclusion only after its coverage of the original task has been proved.

Proof. Original result relations determine the complete profile without reference to the proposal. Soundness and required coverage fail exactly by the differences in Theorem 5.1. Complete decodability fails exactly by the pair and coordinate in Theorem 3.3. A wrong proposed complete interior fails by Theorem 5.4. An unsatisfied certificate condition is a failed original condition at its interpreted locus; when the conclusion is false, Lemma 5.7 localizes a false leaf or non-truth-preserving inference. An unlicensed interpretation cannot be substituted for a transmitted original premise.

Successful realization is the conjunction of these independently interpreted requirements. If the conjunction fails, some requirement fails and the corresponding theorem supplies its witness. Conversely each listed witness contradicts one conjunct. If all conjuncts hold, there is no further failure of those same requirements. This final statement follows from coverage by the original semantics, not from counting the four descriptions of witnesses. A purported additional requirement must either already have its interpreted coordinate or be included and its coverage established. The generated-use result proves such inclusion by the actual formation rules where that method is used. $\square$

Corollary 5.10 (No same-task evasion of a witnessed failure). *A faithful redescription, quotient re-encoding, or new authority label cannot make an incorrect original record correct, fill a missing required record without supplying it, recover a lost original distinction from unchanged sole input, or make an unsafe original continuation safe. A repaired proposal can satisfy the original task only by actually discharging the failed obligation. A changed original task requires its own result or a justified transfer.*

Proof. Faithful transport preserves result differences by Proposition 2.6. Postprocessing h preserves the equality of a colliding pair, so Theorem 3.3 still applies. A faithful path interpretation preserves the node and edge truth values witnessing failure. A label supplies no change to any original semantic condition. Removing a witness therefore requires correction of the proposal, additional justified information, a sound new construction, or a different specified obligation; only the first three can discharge the original task as stated. $\square$

The four witness families are a proof organization, not a claim that reality has four disjoint kinds of failure. Typing, standing answers, quotient information, and continuation locate these witnesses within a proposed realization. Typing and reference claims have original truth conditions, qualification is an original coordinate, quotient adequacy has a fibre test, and continuation has its own contract. No additional governance-bearing claim escapes merely by taking a new syntactic form: once its original meaning is fixed it is tested at that meaning. Conversely a claim whose interpretation or target coverage is absent remains unestablished; it is not licensed by a residual-category argument.

6 History, irreversible retention, and grounded authorization

6.1 History as an original question carrier

Let B be an alphabet of construction acts and B^* its finite histories, with empty word ε and concatenation. Give an independently fixed partial contextual outcome $e : B^* \times B \rightarrow V$. Write $e^\perp$ for its totalization. Define

$$h \simeq h' \iff \forall \sigma \in B^*, a \in B, \quad e^\perp(h\sigma, a) = e^\perp(h'\sigma, a).$$

This includes both definedness and result. Original historical questions that must remain observable, such as the identity of a retained failed trial, must be among the contextual outcomes; future eligibility alone need not remember every historical fact.

Theorem 6.1 (History congruence and complete history factorization). *The relation $\simeq$ is an equivalence stable under right concatenation. It is the largest right congruence preserving all immediate outcomes $e^\perp(h, a)$. Every required history consequence determined by the complete contextual profile factors uniquely through $B^*/\simeq$. Distinct classes have a finite continuation and an original query distinguishing them.*

Proof. Equality of all coordinates is an equivalence. If $h \simeq h'$, then for every ρ, σ, a the words $h\rho\sigma$ and $h'\rho\sigma$ have equal outcomes, so $h\rho \simeq h'\rho$. For any right congruence E preserving immediate outcomes, hEh' implies $h\sigma Eh'\sigma$ for every finite σ , and hence equality of every profile coordinate. Thus $E \subseteq \simeq$. Factorization is the quotient theorem applied to this profile. Inequality of profiles gives a coordinate (σ, a) ; its continuation is finite by its indexing set, not by an assumption of effective evaluation. $\square$

History factorization excludes path dependence only within classes agreeing on every required contextual outcome. If order genuinely affects an original outcome, the corresponding histories occupy different classes. The theorem therefore preserves lawful order dependence while excluding a purported dependence on information the advertised history interface identifies.

6.2 Retention is not endpoint absorption

Theorem 6.2 (Irreversible retention and exact negative-absorption condition). *Let $c : X \rightarrow \{0, 1\}$ be a fixed original qualification and, for a finite retained path, put*

$$H(x_0, \dots, x_n) = 1 \iff c(x_i) = 1 \text{ for all } i \leq n.$$

Extending a retained path preserves every original $c(x_i)$ and cannot change H from 0 to 1. For a relation R , the following are equivalent:

1. *every finite R -path with positive endpoint has $H = 1$;*
2. *every edge xRy satisfies $c(x) = 0 \Rightarrow c(y) = 0$.*

Proof. A path extension does not replace its old entries or their question identities. A previously false conjunct remains false. Apply (i) to a one-edge path to obtain (ii) by contraposition. Conversely, under (ii) any negative entry propagates to the endpoint by finite induction; a positive endpoint therefore has no negative predecessor on that path. $\square$

A bridge repaired after a failed trial can pass a new trial. This does not make the original trial pass. A previously unread proof may already have been valid, while adding a missing premise constructs a different proof. These distinctions retain irreversibility without inferring a universal ban on new qualified outputs. Safe continuation within S does preserve positivity by $S = \Phi(S)$; a negative-absorption claim needs the additional edge condition just proved.

6.3 Composition cannot manufacture an ungrounded warrant

Definition 6.3 (Grounded warrant closure). Let J be a set of interpreted warrant claims, $T \subseteq J$ their original truths, and $B_0 \subseteq T$ independently warranted entrances. A licensed finitary rule is a pair (P, c) with finite premise set $P \subseteq J$ and conclusion $c \in J$, satisfying $P \subseteq T \Rightarrow c \in T$. Independently warranted nullary rules contribute their conclusions to B_0 . Define $C_0 = B_0$ and

$$C_{n+1} = C_n \cup \{c : \text{some licensed } (P, c) \text{ has } P \subseteq C_n\}.$$

Write $\text{Cl}(B_0) = \bigcup_{n \geq 0} C_n$.

Theorem 6.4 (Sound generated authorization and no unsupported entrance). *The set $\text{Cl}(B_0)$ is the least rule-closed set containing B_0 , equals the claims with finite grounded derivations, and is contained in T . If $B_0 = \emptyset$ and every rule has a nonempty premise set, then $\text{Cl}(B_0) = \emptyset$. A cycle of mutually assumed warrants alone supplies no member of this least closure.*

Proof. Induction gives $C_n \subseteq T$ and containment in any closed superset of B_0 . If a finite premise set lies in the union, its members all lie in one C_n by taking the maximum of finitely many stages. Its conclusion therefore lies in C_{n+1} , proving closure and leastness. Induction on stages builds finite derivation trees, while induction on tree height places their conclusions in a stage. When the seed set is empty and premises are nonempty, no rule fires at stage zero, and induction keeps every stage empty. $\square$

This theorem is about least grounded authorization, not the greatest safe interior. The different fixed-point directions answer different questions. A coinductive certificate can be justified by an independently sound coinduction principle; its syntax alone does not establish that principle. Likewise a new constructor theorem can supply a legitimate entrance. It is not an unsupported escape merely because it was absent from an earlier list of proposed warrants.

Corollary 6.5 (Same-task operator closure). *Any finite composition of independently sound warrant transformations preserves original truth. If a composite offered as a grounded certificate has a false conclusion, some original entrance is false or a rule instance fails truth preservation. Introducing additional intermediate names or layers does not remove this alternative.*

Proof. Apply Theorem 6.4 to the finite rule instances of the composition. The false-conclusion alternative is Lemma 5.7. Expanding or naming intermediate nodes does not change their interpreted premises and conclusions. $\square$

6.4 Meta-control as a condition on standing-bearing use

An act claimed eligible for a use must meet that use's original qualification, and an offered certificate must meet its original warrant conditions. This is a logical condition on what the claim can warrant.

It can be realized by checking a record, by a construction whose preservation theorem supplies its warrant, or by a verified combination of these. A temporally separate checker is not required by the semantic statement.

For a procedural implementation, let its state contain a finite committed set $L \subseteq J$. Start with $L \subseteq T$, and permit an append transition only for $j \in T$ established by a sound available warrant. Then every reachable committed set remains a subset of T , by induction on transitions. Candidate formation and evaluation may precede append; an unsuccessful candidate never obtains the claimed positive commitment. This gives a precise implementation invariant without requiring that failed attempts never be evaluated.

The necessary control is therefore at the use of authority: a false original claim does not gain warrant by having been formatted, executed, or logged. If a purported complete authorization system omits an independently warranted entrance, that is a completeness issue. If it admits a false one, that is a soundness issue. Both remain within the exhaustion theorem instead of being concealed by the timing of a gate.

7 Local variation of task and complete profile

A specified interpreted change can alter the original task, alter its complete profile within that task, or preserve both. The following local split names those three effects.

Let $\mathcal{K}$ index original task contracts, with lawful translations already identified or explicitly applied when comparing them. For each κ , let D_κ be the descriptions interpreted in that task and $P_\kappa : D_\kappa \rightarrow B_\kappa$ its complete original profile. Put $D = \bigsqcup_\kappa D_\kappa$. Invalid proposals can be retained as syntactic candidates, but they have no completed original interpretation until their typing is established. An asserted successful interpretation is itself subject to its original typing conditions.

Definition 7.1 (Variation effect). For a specified interpreted change $d \rightsquigarrow d'$:

1. its effect is *anchor* if $\kappa(d) \neq \kappa(d')$;
2. its effect is *tensor* if the task is unchanged and its complete original profile changes;
3. its effect is *skin* if both task and complete profile are unchanged.

A proposed change outside the interpreted domain is an attempted typing/anchor violation when it is asserted as an admissible change within that domain. A lawful transport is compared after its interpretation, as in Definition 2.5, rather than classified as a target change merely because its notation differs.

Theorem 7.2 (Disjoint variation exhaustion and no fourth effect). *Every specified interpreted variation has exactly one of the three effects in Definition 7.1. An effect preserving the original task but changing any required outcome is tensor. An effect preserving every required outcome carries no new authority over those outcomes. There is no fourth original-use effect once P_κ covers the original questions.*

Proof. First split on equality of task indices. Inequality gives anchor and excludes the other two conditions. Under equality, split on equality of complete profiles. Inequality gives tensor and equality gives skin. These cases are disjoint and exhaustive. Since every original outcome is a coordinate of the complete profile, a changed outcome cannot occur in the skin case. Conversely any

alleged further original-use effect has an original question as its meaning; a changed answer at that coordinate changes the profile and is already tensor, unless it changes the task itself. The conclusion uses coverage of original questions, not a stipulation that only three names are available. $\square$

A component's declared variation family can realize more than one of these effects. The local split does not establish an independent factorization into components, a unique decomposition, or minimality of a component system.

8 Consequence determination and uniqueness obligations

Exhaustion determines what counts as a correct realization of an original question. A claim that all compatible candidates have one result requires an additional domain fact. Theorem 8.1 states that fact exactly; Corollary 8.2 distinguishes it from unique occupant identity. Faithful original-task transport remains the result of Proposition 2.6. Full role-based operator identity and component factorization belong to the separate ATS development and supply no premise here.

8.1 One original consequence and a family of candidate occupants

Let R be an independently specified set of compatible original candidates, let $\rho : R \rightarrow Q$ map them to retained quotient data, and let $F : Q \rightarrow Y$ be a specified partial consequence function. A proposed original readout $c : R \rightarrow Y$ satisfies the bridge when

$$\rho(R) \subseteq \text{dom}(F), \quad c = F \circ \rho.$$

These are substantive definedness and interpretation conditions. The function F is single-valued at a fixed input; this alone does not make it constant across $\rho(R)$.

Theorem 8.1 (Exact family-output uniqueness criterion). *Assume the bridge above. The following are equivalent:*

1. *the set of original readouts $c(R)$ is a singleton;*
2. *$R \neq \emptyset$ and F is constant on $\rho(R)$;*
3. *there exist $r_* \in R$ and $y_* \in Y$ such that $F(\rho(r_*)) = y_*$ and $F(\rho(r)) = y_*$ for every $r \in R$.*

In particular, $\rho(R) = \{q_\}$ is sufficient. It is not necessary when distinct quotient inputs have the same required output. If R is nonempty, failure of output uniqueness gives two compatible candidates with different original readouts. If R is empty, uniqueness of a realized output has no existence witness.*

Proof. The bridge gives $c(R) = F(\rho(R))$. This image is a singleton exactly when its domain image is nonempty and all its values agree. Nonemptiness of $\rho(R)$ is equivalent to nonemptiness of R . A selected r_* then supplies its common attained value, proving the equivalences. A singleton input image implies constancy of any defined function on it. Conversely a function can take the same value at different inputs. If the nonempty output image is not a singleton, choose two distinct attained values and their original witnesses. $\square$

Corollary 8.2 (Unique occupants and separating readouts). *Let $R \neq \emptyset$ be the original candidate family, and let $\nu : R \rightarrow \mathcal{O}$ record original occupant identities, with lawful alternative presentations already identified. There is one realized occupant precisely when $\nu(R)$ is a singleton. A singleton readout image $c(R)$ entails that conclusion if c separates occupant identities:*

$$c(r) = c(s) \implies \nu(r) = \nu(s).$$

In particular, an injective readout with singleton image gives a singleton candidate set. A constant readout without the separation condition need not determine a unique occupant.

Proof. The image $\nu(R)$ is exactly the set of realized original identities. Under the separation condition, equal readouts imply equal identities for every pair in the nonempty family. Taking $\nu = \text{id}_R$ gives the injective case. Conversely, a constant function on two distinct identities has a singleton readout image and two occupants. $\square$

Corollary 8.3 (Strong consequence determination export). *For one original input and consequence slot, all sound complete realizations agree. For an original candidate family, a unique realized output is forced exactly when the independently justified readout image is a singleton, equivalently by Theorem 8.1. A new representation or extra authority label cannot change that image under faithful interpretation.*

Proof. The first assertion is Theorem 2.3; the second is the preceding theorem. Faithful recoding commutes with the original readout, so its image changes only by the fixed output bijection. Cardinality one and a separating pair are both preserved. $\square$

This is the distinction needed for downstream numerical and occupant claims. An exact scalar slot specifies what question a scalar answers. Its codomain alone does not prove that the compatible candidates have one value. A domain argument can supply uniqueness by equations, compatibility, rigidity, a complete elimination, or a proved constant-image condition. Once supplied, the present theorem transfers that uniqueness through every faithful complete implementation. It does not replace that domain argument by a prohibition on writing alternative functions.

8.2 Unique normal form versus unique decomposition

Proposition 8.4 (Criterion for decomposition uniqueness). *Let $\mathcal{D}$ be a set of admissible component tuples and $A : \mathcal{D} \rightarrow \mathcal{O}$ their original assembly map into operator identities. An operator o has exactly one admissible decomposition if and only if $A^{-1}(\{o\})$ is a singleton. Existence of one operator identity does not imply this condition. If A is injective and $o \in A(\mathcal{D})$, the decomposition is unique.*

Proof. The fibre is precisely the set of decompositions. Injectivity makes it have at most one member and image membership makes it nonempty. For the failure of the unrestricted implication, let g be a nonidentity bijection with inverse. The same identity operator has decompositions $\text{id} \circ \text{id}$ and $g^{-1} \circ g$ whenever both are permitted. Thus unique composite identity does not identify the tuples. $\square$

A proved product schema, normal form, or reconstruction inverse can establish the required injectivity. The local variation split in Section 7 supplies no such injectivity. The criterion here identifies the additional obligation; the ATS development must establish any role-specific decomposition it claims.

9 Worked constructions and failed claims

The examples pair each claim with a witness and a theorem. They also exhibit successful complete realizations. A catalogue of failures alone would leave the existence and completeness of the admissible construction unestablished.

9.1 A bridge protocol: three behavior classes and one complete interior

Let $X = \{s, t, f, r\}$ describe current bridge configurations. The intact configuration s passes and survives a repeat; the fragile configuration t passes but fails the repeat; f has failed; the repaired configuration r passes and survives subsequent repeats. Set

$$A = \{s, t, r\}, \quad b = 1_A,$$

and declare partial operations

$$t_{\text{repeat}} : s \mapsto s, \quad t \mapsto f, \quad r \mapsto r, \quad t_{\text{repair}} : f \mapsto r.$$

Their domains are exactly the inputs displayed. Every declared edge has its original warrant. This means it is the stated operation on the stated configuration; it does not assert that every edge preserves qualification.

Configuration	b	Repeat result	Repair result	Robust 1_S
s	1	s	unavailable	1
t	1	f	unavailable	0
f	0	unavailable	r	0
r	1	r	unavailable	1

For the original questions consisting of current qualification, operation availability, and their finite continuations, the complete behavior classes are

$$\{s, r\}, \quad \{t\}, \quad \{f\}.$$

The first pair has equal availability and equal outcomes along every word; the repeat distinguishes t from the first class, while the empty-word qualification distinguishes f . Induction on words verifies there are no further separations within $\{s, r\}$. This quotient has three classes although the qualification bit has two. If an additional original question distinguishes the identities or retained trials of s and r , the enlarged profile separates them. The stated three-class calculation concerns the specified behavior questions.

The interior calculation is

$$U_0 = X, \quad U_1 = A, \quad U_2 = \{s, r\} = U_3 = S.$$

Both $\{s\}$ and $\{r\}$ are sound invariant regions, so their union realizes the complete interior. The original failed state has a zero-length failure witness; the fragile state has the one-edge witness $t \rightarrow f$. This verifies both inclusion of every safe state and exclusion of unsafe states.

Four explicit tests.

1. A proposed robust reader reporting $(t, 1)$ has a surplus record. The actual robust graph contains $(t, 0)$. The witness $t \rightarrow f$ supplies its failed local condition. The current-load claim $b(t) = 1$ remains correct; the robust-use question is different.
2. The proposed complete interior $I = \{s\}$ is sound but incomplete. The independently sound region $\{r\}$ supplies the omission witness. A partial certificate reporting only $(s, 1)$ is not thereby false. A total classifier that reports 0 at r additionally supplies an incorrect negative record.
3. The interface $h = b$ has a collision: $b(s) = b(t) = 1$, but the original repeat-success answers are 1 and 0. No sole-input decoder from b can answer that question on both states. The three-class complete behavior quotient supplies the missing distinction and satisfies primitive descent.
4. Substituting the repaired trial for the original failed trial does not discharge the original question. A retained history containing f still has an earlier failure after r is appended. The endpoint can nevertheless be positive, as Theorem 6.2 permits.

These witnesses instantiate graph, interface, continuation, and historical accountability. They do not rely on assigning every case a different failure name. The same fragile configuration can witness an incorrect robust record, an unsafe admission, and a lossy binary interface in different proposals.

9.2 A successful two-class interface

Let $X = \{p_1, p_2, n\}$ with $b(p_1) = b(p_2) = 1$ and $b(n) = 0$. Let one total operation exchange p_1, p_2 and fix n , and let another be the identity with domain $\{p_1, p_2\}$. Take b as the only base observation. Both primitive domains are unions of bit classes; every primitive output has a bit determined by its input bit. The primitive criterion of Theorem 4.3 therefore proves that every finite generated use descends through b .

Both bit values occur, so the complete behavior quotient has exactly two classes. With true edge warrants and local qualification $b = 1$, the complete safe interior is $\{p_1, p_2\}$. It contains two distinct original states and has one positive behavior class. This is a nonempty realization of binary interface closure without an assertion of unique physical occupancy. Adding an original observation distinguishing p_1 and p_2 changes the required complete profile, while their original qualification remains binary.

9.3 Joint compatibility missed by unary inspection

Let each component have two possible identities, 0 and 1, and let its retained unary profile be the constant “qualified.” Suppose the original assembly operation is permitted only on equal pairs:

$$J = \{(0, 0), (1, 1)\} \subseteq \{0, 1\} \times \{0, 1\}.$$

The pairs $(0, 0)$ and $(0, 1)$ have equal retained component tuples, but only the first belongs to J . Thus the joint domain is not saturated and Theorem 4.8 fails with that pair as its witness. Every individual component can pass its local qualification while the proposed composite is impermissible.

Retaining the joint equality observation repairs this specific loss. More generally, retaining identities makes the domain test descend directly. Once every constructor’s joint domain and output conditions

are established, Theorem 4.9 extends the result to every finite assembly tree. The example shows why that theorem checks interfaces between components and not only a list of individually admissible parts.

9.4 Typed construction: an independent safe interior

Consider the simply typed lambda calculus with Boolean constants, conditionals, and types

$$T ::= \text{Bool} \mid T \rightarrow T.$$

Let X be closed raw terms, modulo capture-avoiding renaming of bound variables. Give it the ordinary syntactic call-by-value reduction relation before defining qualification. Applications reduce a function and value argument; a conditional reduces its guard and selects the appropriate branch. Values are Boolean constants and abstractions. At a fixed target type T let

$$A_T = \{t : \emptyset \vdash t : T\}.$$

The original typing rules include

$$\frac{\Gamma \vdash u : U \rightarrow T \quad \Gamma \vdash v : U}{\Gamma \vdash uv : T}, \quad \frac{\Gamma \vdash c : \text{Bool} \quad \Gamma \vdash u : T \quad \Gamma \vdash v : T}{\Gamma \vdash \text{if } c \text{ then } u \text{ else } v : T}.$$

Type preservation establishes $t \in A_T$ and $t \rightarrow t'$ imply $t' \in A_T$. Its beta case uses substitution; its congruence cases use the induction hypothesis; Boolean branch selection uses the corresponding branch typing premise [4, 5]. This is an independently established domain theorem about the syntactic rules, not a definition of reduction as “whatever preserves typing.” With true operational edge warrants, every point of A_T belongs to a sound invariant region, and the zero-length path excludes every point outside it. Hence $S_T = A_T$ by Theorem 5.3. At $T = \text{Bool}$ both Boolean values supply positive witnesses.

Three faulty proposals now have distinct tests.

1. A complete value interface retaining only the typing bit loses the difference between **true** and **false**. Both are typed, while the original question “is this the value **true**?” distinguishes them. This is a fibre collision, not a third typing value.
2. The proposed complete typed interior $J = \{\text{true}\}$ is sound and reduction-closed because the value has no step. It omits the equally sound region $\{\text{false}\}$. This is incomplete coverage, not failure of preservation.
3. A checker that infers original typing from the existence of a typed reduct gives a false answer for

$$u = \text{if true then true else (true true)} \longrightarrow \text{true}.$$

The original term is untypable: its unused application would require a Boolean constant to have an arrow type. Its selected reduct has type **Bool**. The checker’s record $(u, 1)$ is surplus over the original typing graph, whose record is $(u, 0)$. It applies the invalid converse of preservation, although it possesses the syntax needed to evaluate the term.

The original u remains untypable while its new reduct is typed. No original negative fact has been repaired by redescription. Typed preservation can warrant the outputs of a correct constructor without an additional prior checking event. Conversely evaluation success alone is not that preservation theorem. This is the distinction between constructional warrant and an unsupported inference from a later result.

9.5 A uniqueness claim that needs an additional domain theorem

Let $R = \{r_0, r_1\}$, $Q = \{0, 1\}$, $\rho(r_i) = i$, and $F(i) = i$. Every input has one determinate output; all maps are well defined. The original readout family still has two values. No ambiguity at either individual input and no additional selector is required for this multiplicity.

To force one readout, an independent domain condition must eliminate a candidate or prove equal readouts. For example, if a proved compatibility condition restricts the original residue to $\{r_0\}$, Theorem 8.1 gives the unique output 0. Alternatively a proved constant readout $F(0) = F(1) = y_*$ also yields one value while retaining both inputs. These are different domain mechanisms for the same valid export. Specifying the codomain or declaring one slot supplies neither mechanism by itself.

9.6 Finite formation and an eventual-outcome question

On the space of infinite Boolean traces, the question “does a 1 eventually occur?” has a determinate classical truth value for each complete trace. Every finite all-zero prefix can be extended to a trace with a later 1 or to the all-zero trace. Thus no fixed finite prefix determines this answer for all traces. The complete original profile can retain the eventual-outcome coordinate directly. A finite-prefix interface fails its fibre test, and a grammar restricted to finite inspected prefixes does not establish coverage merely by naming the question.

The example preserves the necessity of admissibility and reference for the trace question while identifying the actual information obligation. Exhaustion remains available at the original semantic level; an effective finite decision procedure is a separate and here impossible claim for that interface.

10 The dependency theorem and its downstream use

Theorem 10.1 (Canonical fixed-domain closure package). *Every independently specified determinate-use task with set-sized original questions has the following canonical structure and exact realization tests.*

1. *Its complete outcome profile determines a unique minimal complete quotient up to the unique commuting bijection. Distinct classes have a separating original question.*
2. *Every complete faithful representation determines that quotient. A deficient proposed representation has a same-input/different-outcome witness. Primitive and joint constructor descent extend to all finite generated uses; coverage of the original practice is established by its interpretation in those formation rules.*
3. *Each original qualification is bivalent, and complete sound realizations of that qualification agree. The whole interface is binary exactly under Theorem 3.6.*
4. *Each declared local-safety/reuse contract has a unique greatest complete safe interior. Unsound inclusion has a finite unsafe-path witness; incomplete exclusion has an omitted sound-region witness.*
5. *Every semantic failure of a proposed complete realization has an original result, information, continuation, or support witness as in Theorem 5.9. Faithful redescription preserves the original discrepancy. Unchanged-input postprocessing cannot resolve a witnessed information collision.*

6. *Original consequences and retained historical claims are conserved at their original incidences. Grounded compositions preserve sound authorization; unsupported cycles do not supply their own entrance.*
7. *A claimed unique family output, original occupant, or component decomposition requires the corresponding image, identity-separation, or assembly-fibre condition in Section 8. The local variation split describes task/profile changes; it does not establish an ATS factorization or an operator-identity theorem.*

All determinate objects and uses appearing in these constructions incur the kernel roles necessarily. The mathematical representation tests specify how their original questions are faithfully realized.

Proof. Part (i) is Theorem 3.2 and Corollary 3.4. Part (ii) follows from Theorem 3.3, Theorems 4.3 and 4.5, Corollary 4.10, and Theorems 4.8–4.9. Part (iii) is Theorem 2.3 and Theorem 3.6. Part (iv) is Theorems 5.3 and 5.4. Part (v) is Theorem 5.9 and Corollary 5.10. Part (vi) is original-incidence agreement, Theorems 6.1 and 6.2, and Theorem 6.4. Part (vii) is Theorem 8.1, Corollary 8.2, and Proposition 8.4; the local split is Theorem 7.2. The final constitutive statement concerns the actual determinate bearers, rather than an optional adoption of their representations [6]. $\square$

10.1 What a later paper can import

The following table names the available result, the proof needed to apply it, and a conclusion it does not transfer. All numbering is internal to this edition. The table separates mathematical application conditions from necessary kernel governance of the original determinate objects.

Invocable result	Required application proof	What is not transferred
Complete interface: Theorems 3.2–3.3	Include every original question and semantic domain; prove fibre constancy.	Correctness of a proposed reader; adequacy of an omitted question.
Finite generated coverage: Theorems 4.3, 4.5; Corollary 4.10	Prove the continuation-profile criterion and original-use interpretations; primitive descent is a sufficient route.	Coverage of eventual success merely from finite-prefix coverage.
Joint assembly: Theorems 4.8–4.9	Saturated joint domains, retained output agreement, and strict finite-tree interpretation.	Joint compatibility from unary qualification alone; an ATS decomposition.
Exhaustion: Theorem 5.9	Independently fixed original graphs, interface claims, reuse and certificate obligations.	A new physical law, numerical value, or correctness certificate from a taxonomy.
Safe interior: Theorems 5.3–5.4	Independent R, A, W and complete coverage of the declared reuse contract.	Nonempty realization or one occupant from greatestness alone.

Invocable result	Required application proof	What is not transferred
Original answer: Theorem 2.3; transport: Proposition 2.6	The same original question; soundness and required coverage; commuting faithful interpretation.	Equality of different consequences, mechanisms, or candidate outputs.
History and warrant: Theorems 6.1, 6.2, 6.4	All required contextual outcomes; retained original incidences; true entrances and sound rules.	Erasure of an old failure; endpoint absorption without its edge condition; ungrounded authority.
Family output: Theorem 8.1	The bridge $c = F\rho$, definedness, nonempty R , and constancy on $\rho(R)$.	A singleton output from the name or type of its slot.
Occupant: Corollary 8.2	A singleton original identity image; a singleton readout must also separate identities.	One occupant merely because several occupants give the same scalar.
Decomposition criterion: Proposition 8.4	A nonempty singleton assembly fibre, proved by the application.	Exclusive ATS components, minimality, or unique factors from one composite.

A downstream application fixes its bearer, original question, incidence, and input/output meanings before applying these results. It proves that its interface covers those requirements, including definedness, joint compatibility, and relevant history. It then verifies the claimed result graph, safe interior, or certificate conditions. An additional numerical, occupant, or decomposition conclusion requires its corresponding substantive domain proof. These obligations concern what the proposed realization establishes about an already determinate target.

10.2 Closure by exhaustion and impossibility

Exhaustion establishes the following impossibility results. A witnessed original contradiction does not become a correct result through relabeling. A lost distinction is not recovered by a new name for the same sole input. A false certificate is not validated by nesting it in a larger unsupported certificate. An unsafe admitted configuration remains unsafe for the same declared repertoire. An omitted safe configuration defeats a claim to have exhausted that repertoire. Each impossibility has a stated subject and a witness that survives faithful interpretation.

At the same time, a correction can discharge an obligation, a new warranted observation can resolve an information deficit, and an independently sound constructor can produce a new qualified object. None of these changes the original failed occurrence into a successful one. Exhaustion therefore constrains the authority claimed for constructions without prohibiting actual construction.

10.3 Scope of the completed argument

Three kinds of work are connected but not identified. The kernel establishes necessary functional conditions of determinate objecthood. The present mathematical theorems establish canonical profiles, complete realization criteria, and exhaustive failure alternatives. Particular mathematical and physical applications establish their original predicates, interpretations, compatible candidates, and substantive consequences. The theorem chain preserves their dependency direction.

The proof methods are explicit: quotient universality, induction on finite words and constructor trees, fixed-point characterization of local safety, and localization in well-founded certificate dependencies.

The theorem supplies no universal algorithm for deciding all semantic questions. Its project includes bounded finite-model checks of the mathematical alternatives; these corroborate instances and do not replace the general proofs. The full manuscript is not represented as a completed proof-assistant formalization.

The decisive external tests remain precise: an original use omitted by the claimed coverage; a pair defeating an asserted factorization; a valid same-task realization outside a claimed complete classification; a sound region omitted by a claimed complete interior; a failed inference; or an unproved uniqueness bridge in a downstream application. Relabeling an adequate realization is agreement with the theorem. Exhibiting a genuinely missed original requirement or an invalid proof step is a substantive challenge. This is the sense in which the fixed-domain result supports the corpus: closure is earned by complete interpretation and exact alternatives, and remains inspectable at every downstream use.

A Dependency order and the total-update universal property

A.1 An acyclic proof order

The original result relations precede their proposed realizations. The profile precedes the quotient; quotient adequacy precedes any proposed compression. Primitive descent precedes finite-word and constructor induction. The safe interior uses independently fixed node and edge conditions. Exhaustion then combines these established alternatives. The local variation split and uniqueness applications supply no premise to the earlier profile or coverage proofs. No ATS component or operator theorem is imported.

Result family	Actual mathematical inputs
Original agreement	Single-valued original result relation; soundness and stated coverage.
Complete profile	Set-sized original question family; uniquely totalized partial results.
Quotient and decoder	Equality of original profiles; fibre constancy; attained images.
Binary reduction	Qualification as one coordinate; constancy of every required coordinate on its bit fibres.
Continuation congruence	Deterministic partial operations; separate domains; all finite continuation coordinates.
Generated observation coverage	The declared finite expression grammar; original interpretation identities where a further target is claimed.
Greatest safe interior	Independent R, A, W ; universal safety of finite paths.
Result family	Actual mathematical inputs
Graph and interior exhaustion	Both directional differences; greatestness and path semantics.
Certificate localization	Original interpretations and well-founded dependencies.
Grounded authorization	True entrances; truth-preserving licensed finitary rules.
Local variation split	Fixed task indices; complete profiles; a specified variation as the unit.
Joint assembly	Saturated joint domains; preserved retained outputs; finite constructor trees.
Family-output uniqueness	A nonempty original candidate family, readout bridge, and constancy on its image.
Unique occupant	Original identity map; nonempty singleton image, or a singleton readout separating identities.

These inputs do not require that an arbitrary candidate be correct, that every invariant be a function of one bit, that every safe interior be inhabited, or that every original question be decidable. The kernel-governance conclusion concerns the determinate objects of the construction; the table lists the mathematical hypotheses of their specific representation results.

A.2 The categorical form of the original quotient spine

For completeness, take a monoid U acting by total maps on X and a family of total observations $o_j : X \rightarrow V_j$. Define

$$x \sim y \iff o_j(u(x)) = o_j(u(y)) \text{ for all } u \in U, j.$$

This is the largest U -stable equivalence preserving present observations, by the same argument as Theorem 4.2. Let $Q = X/\sim$ and $q : X \rightarrow Q$.

An observed U -system consists of a U -action on a set Y and observations $\tau_j : Y \rightarrow V_j$. A morphism $f : Y \rightarrow Z$ commutes with every action and observation. Identity functions and function composition satisfy these conditions, so they form a category. Let

$$E = \{(x, y) \in X \times X : x \sim y\}.$$

It is an observed U -system with diagonal action and observations $o_j(x) = o_j(y)$. The equality uses the identity element of U ; stability makes the action remain in E . Both projections $r_1, r_2 : E \rightarrow X$ are morphisms.

Theorem A.1 (Observed-system coequalizer). *The quotient has the well-defined action $\bar{u}([x]) = [u(x)]$ and observations $\bar{o}_j([x]) = o_j(x)$. Its projection q is a morphism and is a coequalizer of r_1, r_2 : for every observed-system morphism $f : X \rightarrow Y$ with $fr_1 = fr_2$, there is a unique morphism $\bar{f} : Q \rightarrow Y$ with $f = \bar{f}q$.*

Proof. Stability gives well-defined action and its identity/composition laws descend from those on X . Observation preservation gives well-defined $\bar{o}_j$. The two preservation equations for q hold by definition, and $qr_1 = qr_2$ because pairs in E are equivalent.

The equalizing condition on f says it is constant on $\sim$ -classes, so put $\bar{f}([x]) = f(x)$. For each u , using equivariance of f gives

$$\bar{f}(\bar{u}([x])) = f(u(x)) = u_Y(f(x)) = u_Y(\bar{f}([x])).$$

For each observation, $\tau_j(\bar{f}([x])) = \tau_j(f(x)) = o_j(x) = \bar{o}_j([x])$. Thus $\bar{f}$ is a morphism; surjectivity of q gives uniqueness. $\square$

This preserves the universal property of the total-update presentation. The equalizing condition is material: not every arbitrary map on raw descriptions factors through q . Partial updates require domain saturation in addition, as proved in Section 4.

Corollary A.2 (No loss of future distinctions in a complete present quotient). *Suppose x, y agree on present observations but some $o_j(u(x)) \neq o_j(u(y))$. No equivalence identifying x, y can both be stable under every update and preserve every present observation.*

Proof. Stability would identify $u(x), u(y)$; observation preservation would equate their o_j values, contradicting the given difference. $\square$

This is a mathematical obstruction before any claim about temporal implementation. Irreversible retention additionally forbids changing the attributed original occurrence merely to conceal that witnessed ambiguity.

B Separating models and limits of stronger inferences

The following models identify which conditions do real work. They are not presented as independence of a large bundled axiom system; each isolates the named implication. Exact equivalences in the main text already establish necessity and sufficiency for their stated tasks.

Observation coverage. Let $X = \{0, 1\}$, let the admitted update be identity, and let the selected observation family contain only a constant map. Its quotient has one class. The independently required observation $x \mapsto x$ does not descend. A list of tests is therefore not complete merely because it was declared. Adding the required observation, or proving it generated, supplies the missing bridge.

Qualification versus complete behavior. Let every state have qualification 1 while an original identity observation separates them. The qualification quotient has one class; the complete quotient can have any cardinality. Conversely a carrier can be empty, in which case its quotient has no classes. The exact two-class assertion requires both bit values and complete-profile factorization through that bit.

Observation definedness. Let $h(0) = h(1)$, and let an observation be defined only at 0. Its defined value need not disagree with any other value, but its partial domain already fails to descend. Omitting the absence/definedness coordinate loses an original distinction.

Operation domain saturation. Let $h(0) = h(1)$, all observations be constant, and an operation have domain $\{0\}$. No partial operation on the sole quotient class has the correct availability at both inputs. Output agreement on the original domain alone does not repair this failure.

Primitive output congruence. Let $X = \{x, y, u, v\}$ with constant observations and one total update $x \mapsto u, y \mapsto v$, fixing u, v . An interface merging x, y but separating u, v retains every generated observation while failing update descent. The distinction between observational adequacy and preservation of an enriched interface's own output classes is therefore necessary.

Joint domains. Two constant component profiles cannot determine equality of hidden component identifiers. The diagonal domain in Section 9 is not saturated on their product fibres. Local unary success does not replace the joint constructor test.

A false invariant answer. On $X = \{x\}$ let the original Boolean answer be 0 and the proposal say 1. Both are invariant under every identity redescription. The proposal still has the surplus record $(x, 1)$. This demonstrates why graph correctness is checked separately from factorization.

Sound but incomplete coverage. With two qualified fixed states, each singleton is a sound invariant region, while their union is the unique complete interior. Reporting only one true positive is sound as a partial practice. Calling it complete adds a false claim. A total negative report on the omitted state is additionally incorrect.

Historical retention and a positive new endpoint. The warranted edges $f \rightarrow r$ and $r \rightarrow r$, with $c(f) = 0$ and $c(r) = 1$, give a positive repaired endpoint while retaining a failed earlier history. Negative absorption is therefore an additional transition property, not a consequence of immutable original records.

Quotient factorization versus output uniqueness. With $q = \text{id}$ on $\{0, 1\}$, both $F(x) = x$ and $G(x) = 1 - x$ factor. If nontrivial fibres are desired, use $q(i, r) = i$ on $\{0, 1\} \times \{a, b\}$. Both functions still descend. Equality of all consequences cannot be inferred from the universal property for each given consequence.

Variation effects versus coarse components. A component with valid replacements 0, 1 and an invalid replacement can expose both an interior consequence-changing direction and a typing violation. It is not exclusively typed by the existence of either one. For interpreted effects, choose descriptions (κ, p, s) : changing κ , changing p at fixed κ , and changing only s realize the three distinct cases. This witnesses nonempty effect categories without making their labels mandatory terminology.

Composite identity versus decomposition identity. For an involution $g \neq \text{id}$, both $\text{id} \circ \text{id}$ and $g \circ g$ equal id . Injectivity of an assembly map is a separate condition from equality of its composite outputs.

Ungrounded cycles. The rules $a \Rightarrow b$ and $b \Rightarrow a$ with no entrance have empty least grounded closure. The set $\{a, b\}$ is nevertheless closed under the rules. Thus being a closed set of mutually assumed warrants is not being the least closure generated by independently warranted entrances. Greatest fixed-point reasoning has valid applications, including safe interiors, but cannot be substituted for the warrant question without its own sound interpretation.

The finite checks in the project compare independent constructions of behavior equivalence, safe regions, and graph alternatives. Their role is to detect small mistakes and confirm the displayed separating models. The general claims rest on the proofs, and actual domain coverage rests on the original interpretation.

C Source dependencies and downstream applications

The operative export table is in Section 10. The division of labor in Section 1 leaves the general original-task, profile, coverage, witness, history, and composition proofs in this paper. The separate ATS development concerns component-role factorization, minimality, and operator identity. It may import the present generic constructor theorems; the present proofs do not import those later role conclusions.

The local task/profile split of Definition 7.1 and Theorem 7.2 classifies a specified variation. A product decomposition of descriptions, independent variation of components, and unique component typing each require their own hypotheses and proofs. The component-role and operator-identity results are developed in the ATS manuscript [12].

C.1 Source dependencies

Finite generator verification is explicitly used in CNF1, Lemma 8.5 [9]. Variation-level classification and joint effects occur in *Standing Carrier Identity Closure*, Theorems 5.3–5.4 [10]. Consequence-complete witness bases and restricted-data-access arguments occur in *Constraint Architectures for Coherent Evaluability*, Definition 2.2 and Lemma 2.10 [11]. These are useful instances and motivations for the constructor and coverage problems.

Their global closure claims are not premises of the present exhaustion proof. Several such corpus developments themselves import fixed-domain exhaustion. The mathematical induction, fibre, and variation arguments required here have therefore been proved locally, with their original data and formation rules explicit. The dependency order is therefore acyclic: downstream applications may invoke the present theorem chain once its stated application conditions have been established.

C.2 A reusable statement of a downstream application

A precise application can take the following form. Let $\mathcal{T}$ be the original target, let $\mathcal{Q}$ include all its required questions, and let h be the proposed interface. Establish the original interpretation and show that each required question is determined by h , either directly or through the generated-use theorem. Verify the result graph, any claimed safe interior, and the original certificate conditions. Theorem 5.9 then excludes every semantic failure of that complete realization. Corollary 5.10 shows that an independently witnessed failure cannot be repaired by a faithful change of presentation alone.

For a uniqueness conclusion across an original candidate family, additionally give R, ρ, F and the proved bridge of Theorem 8.1, together with nonemptiness and constancy. For a unique component decomposition, provide the singleton assembly fibre of Proposition 8.4. For a physical or mathematical application, these are the substantive domain premises to which the universal theorem is applied. These premises determine which uniqueness conclusion follows in the application.

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